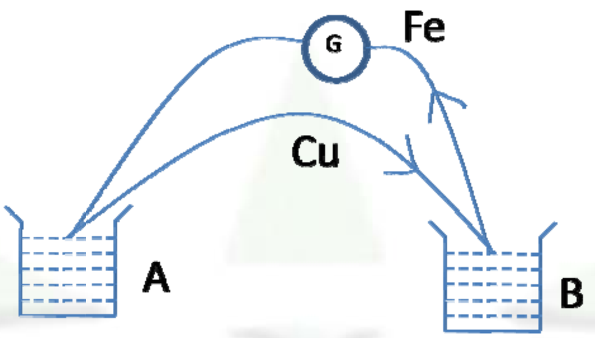
Thermo electric effect:

Seebeck Effect: When the junctions of two dissimilar metals are kept at two different temperature, it is found that a current flows through the circuit. This phenomenon is known as Seebeck effect or thermoelectric effect. The current flowing through the circuit is known as thermo electric current.



The emf responsible for this current is known as Seebeck emf and the circuit of two dissimilar metals is known as thermo couple.

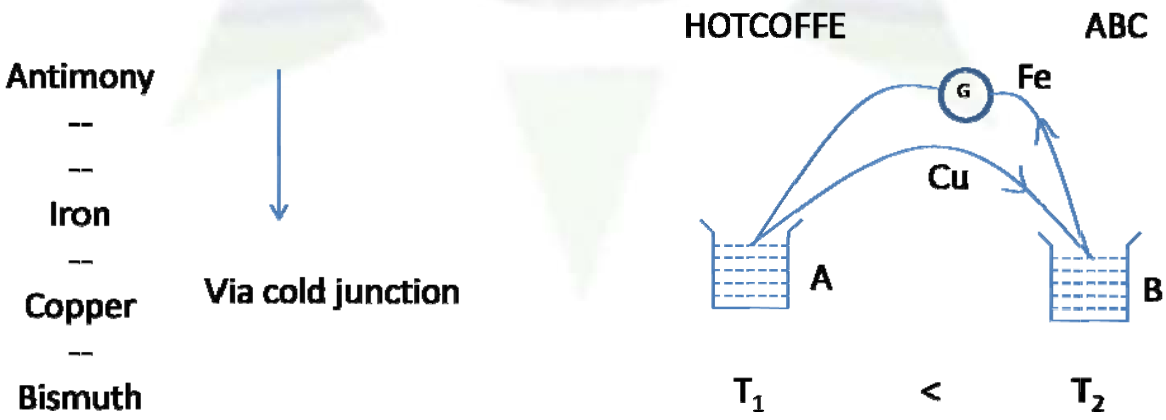
The thermo emf produced in thermo couple depends on two factors:

(1) The elements forming the thermocouple

(2) The temperature difference between the two junction

Generally thermo emf varies from 2μv/^{0}C to 30μv/^{0}C

Seebeck arranged all the element in a table maintaining sequence. The thermo electric current flows from the elements occupying higher position in the table to the lower position via the cold junction. The table starts with antimony and ends with bismuth.

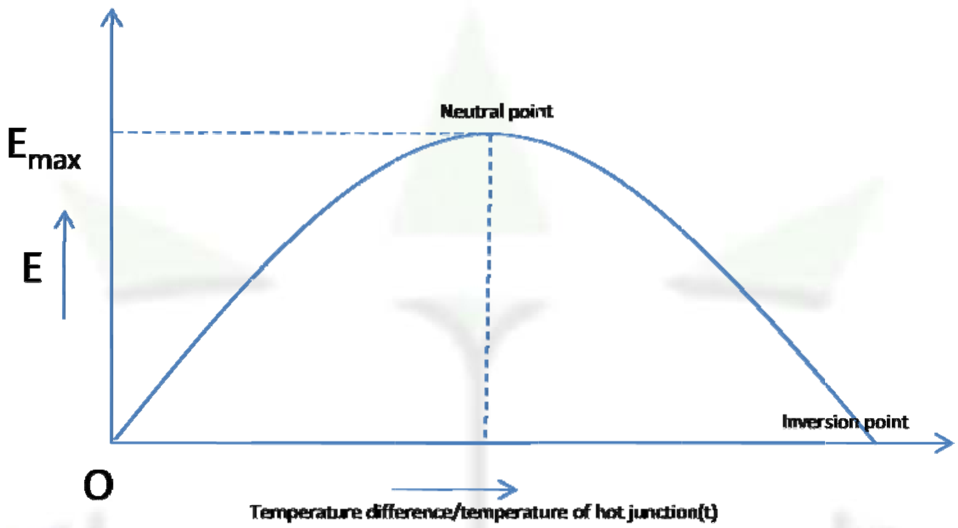


ABC = Antimony to copper through cold junction

HOTCOFFE = Copper to Iron via hot junction

Variation of thermo emf with temperature difference:

Keeping the temperature of the cold junction constant if the temperature of the hot junction is increased the temperature difference increases. Experimentally it is found that as the temperature difference increases the Seebeck emf or themo emf also increases reaches a maximum value at a particular temperature of the hot junction known as neutral temperature T_n.



As the temperature is increased further the thermo emf decreases and becomes zero. If the temperature is increased still further the thermo emf starts increasing again but in an opposite direction. The temperature at which the themo emf becomes zero and reverses its sign is known as temperature of inversion.

It is found that the neutral point is a constant of thermo couple and does not depend on the temperature of the cold junction. But the inversion temperature is not a constant for a given thermo couple and its value depends on the temperature of the cold junction.

The inversion temperature T_i is always as high above the neutral temperature as the cold junction below it.

$$T_i - T_n = T_n - T_0$$

$$T_n = \frac{T_0 + T_i}{2}$$

For example for Cu-Fe thermo couple the neutral point T_n=550^{0}C

T_0	T_n	T_i
0	550	1100
100	550	1000
250	550	850

Explanation of Seebeck effect: According to the electron theory metals contain free electrons behaving like a perfect gas. The pressure and density of electron gas depend on the temperature and material of the body. When two dissimilar metals are joined to form a thermocouple electron diffuse from the metal of higher density to the one at lower density, producing a dis-balance of positive and negative charges in the metals at the junctions. An electric field is set at each of the junctions which builds up as more and more electrons diffuse from one to other. When the electric field force on the electrons is sufficient to prevent a further diffusion of electrons a constant electric field is set at the junctions. Thus the two junctions may be represented as the tiny cells. Normally the temperatures of the junctions being the same the emfs of these two tiny cells are equal and opposite and hence no current is produced in the circuit. But when the junctions are at different temperatures the emfs of those tiny cells are unequal and hence they give rise to a resultant emf which maintains the flow of charge carriers in the circuit.